

Theoretical Assessment Questions for the 2016 International

Biology Olympiad

National Team Selection **Camp**

Student Code



Precautions:

1. This test paper consists of 5 volumes.

The test will last from 8:30 to 12:00, for a total of 210 minutes.

2. This test consists of five parts, each worth 100 points, for a total of 500 points.

3. When answering the questions, please check if the cover of the test paper matches your test number.

4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.

5. You may write your answers on the back of the questions, but please indicate the question number.

Part One: (100 points)

PART A: Sorting questions (only full marks for correct answers), fill-in-the-blank questions (40 marks total)

1. During each annual influenza epidemic, the Centers for Disease Control and Prevention (CDC) of the Ministry of Health and Welfare identifies the pathogens.

Please rank the following identification methods in order of accuracy, with the most accurate one listed first. **(3 points)**

- (A) Changes in a patient's body temperature
- (B) Detection of serum antibodies
- (C) Types of clinical symptoms
- (D) Viral nucleotide sequencing
- (E) Analysis of membrane glycoprotein subtypes

_____ ^y _____ ^y _____ ^y _____ ^y _____

2. Leukocyte extravasation includes the following steps. Please arrange them in order. **(3 points)**

- (A) Integrin activation
- (B) Rolling
- (C) Binding to cell adhesion molecules
- (D) Transendothelial migration
- (E) Adhesion

_____ ^y _____ ^y _____ ^y _____ ^y _____

3. Botulinum toxin (BTX) is one of the most potent natural toxins, derived from *Clostridium botulinum*.

Please arrange the process from the formation of botulinum toxin to the point of causing food poisoning in sequence. **(3 points)**

- (A) Cell death releases protoxins
- (B) *Clostridium botulinum* cell proliferation
- (C) Intracellular synthesis of protoxins
- (D) Blocking acetylcholine secretion
- (E) Protease-activated protoxin

_____ ^y _____ ^y _____ ^y _____ ^y _____

4. Cholera is caused by *Vibrio cholerae* and can lead to severe diarrhea, vomiting, dehydration, acidosis, and even shock.

Count. Please arrange the following processes in chronological order of occurrence. (3 points)

(A) Bacteria multiply in the epithelial cells of the small intestine and produce

enterotoxins. (B) The alpha units of the enterotoxins enter the cells.

(C) Activation of adenylate cyclase leads to ATP $\rightarrow$ cAMP

(D) The γ -unit of enterotoxin binds to receptors on the surface of epithelial cells, causing allosteric changes in the toxin molecule.

(E) Increased intracellular cAMP levels stimulate the intestinal mucosal cells to secrete large amounts of cAMP.

_____ γ _____ γ _____ γ _____ γ _____

5. The following intermediates are produced in the classical pathway of complement activation. Please list them accordingly.

Arrange the intermediate products in the following order. (3 points)

(A) C5b

(B) C1

(C) C4b2a3b

(D) C4b

(E) C2b

_____ γ _____ γ _____ γ _____ γ _____

6. Epithelial cadherin (E-cadherin) is present on all epidermal cells, and its main function is to create intercellular connections.

E-cadherin in epithelial cells is essential for maintaining the integrity of tissue structure. Abnormalities in E-cadherin levels can lead to...

Cellular carcinogenesis. Young scientist Brendan recently isolated a cell line he believes can secrete E-cadherin.

Specific antibodies can block the adhesion between mouse embryonic stem cells. He went through certain...

The experimental procedure involved using this antibody to identify the cDNA encoding E-cadherin and demonstrating that this cDNA truly...

It can encode a cell adhesion molecule. Please arrange his experimental steps in chronological order. (5 points)

(A) Generate DNA using reverse transcriptase. (B) Screen

for colonies that express the target protein of this antibody.

(C) Expression of this antibody in non-adhesive cell lines induces cell adhesion, but it can also block cell adhesion.

(D) Using plasmids as cDNA gene banks

(E) mRNA isolated from adhesion cells

_____ γ _____ γ _____ γ _____ γ _____

7. Suppose that a specific signal transduction pathway in a cell requires the following five proteins:

- A) Ca⁺⁺ channel
- B) transcription factor
- C) receptor
- D) G-protein
- E) phospholipase C

a) When this specific signal reaches this cell, please write down the order in which it is transmitted (5 points, only complete correct answers will receive a full score).

point)

_____ y _____ y _____ y _____ y _____

8. Following on from question 7 in the sequencing section, where are these five proteins most likely located in the cell? (Fill in the blank)

(15 points, 3 points per square)

A	
B	
C	
D	
AND	

PART B: Short Answer Questions (13 points total)

1. In the modern world, there are far more enzymes producing proteins than ribozymes producing RNA. This is because...

Proteins have strong enzyme specificity, fast catalytic rate, and a wide range of catalytic reactions.

Is this reasoning correct? Why? (8 **points**)

2. Lipid-soluble signaling molecules, such as testosterone, can cross the cell membranes of almost all cells.

However, only a small percentage of cells can receive and generate the signal. What is the main reason for this? (5 **points**)

PART C: Fill in the blanks (34 points)

The cytoskeleton refers to the fibrous network structure of proteins in the cytoplasm; it belongs to...

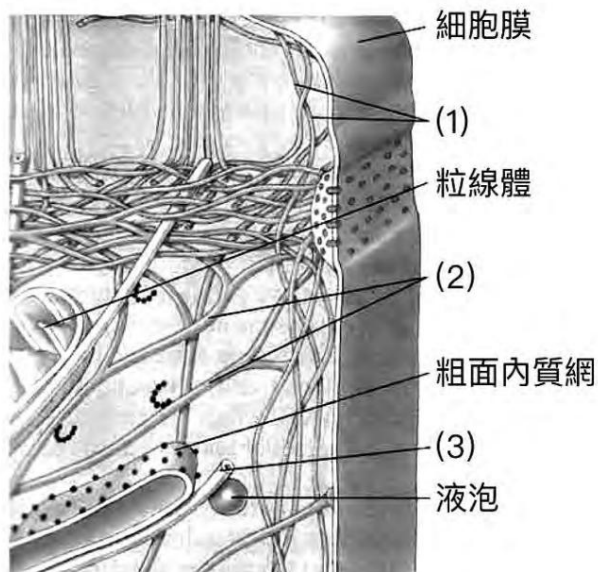
A dynamic structure, meaning that a part of it is constantly being destroyed, renewed, or newly formed. (This applies to living organisms.)

The cytoskeleton system is composed of a group of similar proteins; due to differences in biological morphology and cell type...

The cytoskeleton is quite high, therefore, the structure, function, and dynamic behavior of the cytoskeleton can vary.

1. The following is a cartoon diagram of a eukaryotic cell structure. Please fill in the blanks in the diagram labeled (1), (2), and (3).

Choose the correct cytoskeleton name from the options (A, B, and C). (3 **points**)



3 points for 3 correct answers; 1 point for 2 correct answers; 0 points for 1 or 0 correct answers.

Answer	Question Number	Options	
	(1)	(A)	Microtubule
	(2)	(B)	Microfilament
	(3)	(C)	Intermediate filament

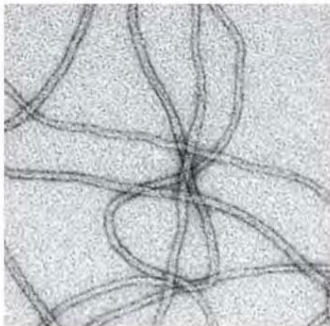
2. The image below shows a transmission electron microscope image of a eukaryotic cell cytoskeleton.

The results of microscopy. Please fill in the corresponding positive values between the questions marked (4), (5), and (6) in the figure.

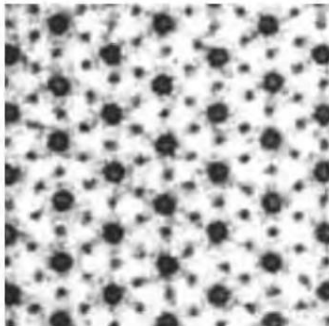
Please select the correct cytoskeleton name from the options (A, B, and C) (3 points).

3 points for 3 correct answers; 1 point for 2 correct answers; 0 points for 1 or 0 correct answers.

Answer	Question Number	Options	
	(4)	(A)	Microtubule
	(5)	(B)	Microfilament
	(6)	(C)	Intermediate filament



(4)



(5)



(6)

3. The cytoskeleton plays a vital role not only in maintaining cell shape, withstanding external forces, and preserving internal cell structure, but also in many important life activities. Please fill in the correct life phenomena (A, B, and C) corresponding to the three main cytoskeleton components (labeled as (7), (8), and (9)). (3 points)

3 points for 3 correct answers; 1 point for 2 correct answers; 0 points for 1 or 0 correct answers.

Answer	Question Number	Options
	(7) Microtubule	(A) Muscle cell contraction
	(8) Microfilament	(B) Chromosome separation during mitosis
	(9) Intermediate fiber filament)	(C) Small intestinal endothelial cell villi support

4. The cytoplasm of an amoeba can be divided into two layers, an inner and an outer layer, which exist in different states. The outer layer is thinner, contains no particles, and is fluid-like, easily flowing; it is called cytosol. The inner layer, protoplasm, is thicker, contains various organelles, and is gel-like, not easily flowing; it is called cytogel. In the outer cytosol, alpha-tubulin and beta-tubulin exist in unit form, so the viscosity is relatively low and the fluidity is relatively high. In the inner cytogel, alpha-tubulin and beta-tubulin form aggregates, so the viscosity is relatively high and the fluidity is relatively low. Please select from the following options the drugs, substances, or physical environments that can hinder the movement of amoebas. Mark 'O' for correct answers and 'X' for incorrect answers. (4 points)

Scoring method: 4 points for 4 correct answers; 2 points for 3 correct answers; 1 point for 2 correct answers; 0 points for 1 or 0 correct answers.

Answer	Question Number	Options
	(10) Colchicine	
	(11) High calcium solution	
	(12) Paclitaxel	
	(13) Low temperature environment	

5. The image below shows HeLa cells undergoing division. Figure (A) shows the cells after staining with two different fluorescent dyes, and the results are observed separately.

The result of taking a picture and then overlaying it; Figure (B) is DAPI (4',6-diamidiny-2-phenylindole (4',

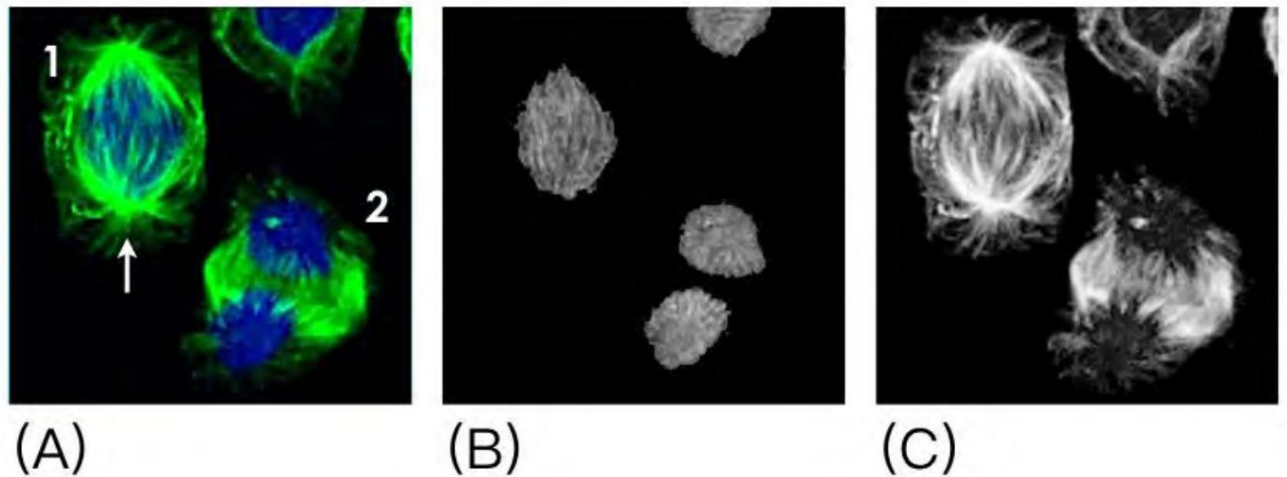
The staining results of 6-diamidino-2-phenylindole (DAPI) show that DAPI is a strong DNA binder.

The combined blue fluorescent dye; Figure (C) shows the green fluorescent dye.

5.1. Please answer the following questions (14-16): What is the cytoskeleton marked with a green fluorescent line in Figure (C)?

In the following options, mark 'O' for correct and 'X' for incorrect. (3 points)

3 points for 3 correct answers; 1 point for 2 correct answers; 0 points for 1 or 0 correct answers.



Answer	Question Number	Options
	(14)	(A) Microtubule
	(15)	(B) Microfilament
	(16)	(C) Intermediate filament

5.2. (17) What structure does the arrow point to? (3 points)

5.3. Cell cycle refers to the period from the end of one mitotic division in a continuously dividing eukaryotic cell to the next cell cycle.

The cycle of growth until the end of the next division. In Figure (A), which cells are cells 1 and 2 (question number...)?

(18-19) In which stage of the cell cycle are they respectively? (2 points)

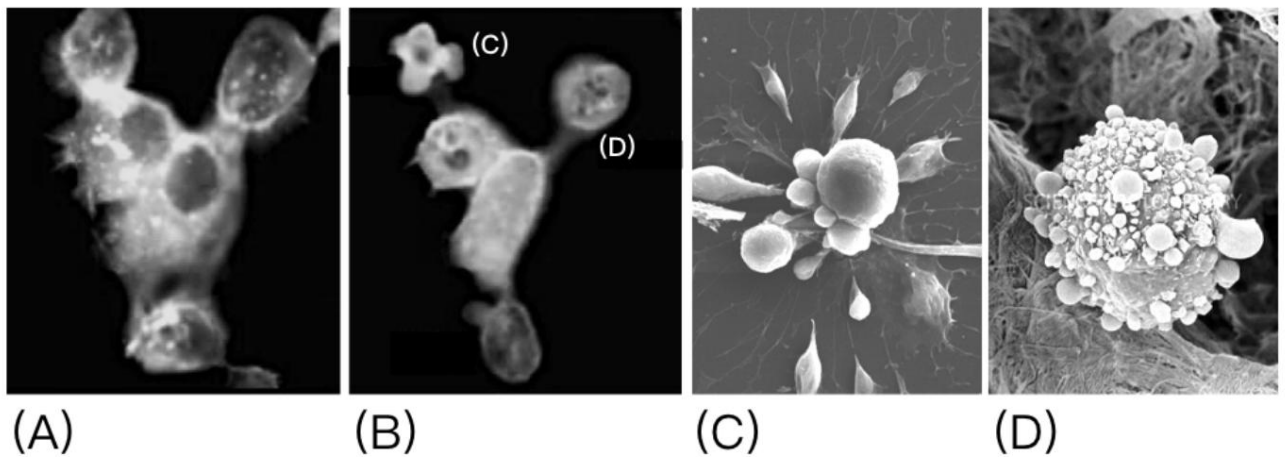
Answer	Question Number	Options
	(18)	(A) G1
	(19)	(B) S
		(C) G2
		(D) M

6. Using flow cytometry, immunofluorescence, or immunochemical staining, researchers can quickly and accurately determine the nature of tissues.

The cell cycle state or tissue location of a single cell within a population of proliferating cells. The cell cycle is generally divided into four phases.

The cell cycle consists of four phases: G1, S, G2, and M. The amount of cellular DNA differs in each phase.

Therefore, the cell cycle can be determined by the amount of DNA in the cell.



6.1. **Figure** (A) shows HeLa cells cultured normally, and Figure (B) shows cells cultured for 48 hours after the addition of OBI-822.

Fluorescence images of actin. Figures (C) and (D) are labeled (C) and (D) in Figure (B), respectively.

The results of scanning electron microscopy.

Please answer the following questions (20-21): The cells in Figures (C) and (D) die due to the addition of OBI-822.

Please indicate which situation each belongs to. Mark 'O' for correct and 'X' for incorrect in the following options.

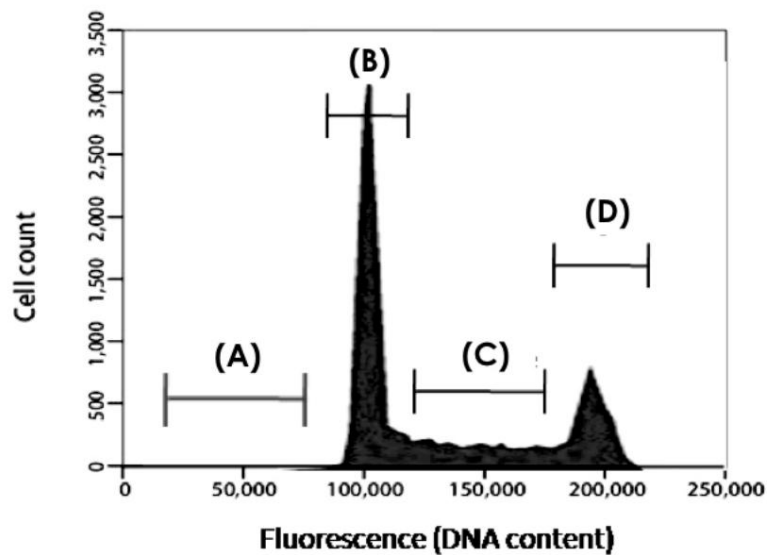
(4 points)

Question Number	Apoptosis (apoptosis)	Cell necrosis (necrosis)
(20)		
(21)		

6.2. The following figure shows the results of flow cytometry applied to cell cycle analysis.

Please answer question 6.1. In which region would the cells in Figures (C) and (D) (questions 22-23) be located?

The correct answers (A ~ D). (4 points)



Question Number		Answer
(20)	Figure (C)	
(21)	Figure (D)	

7. Molecules, organelles, and reactions inside the cell membrane are important; it was previously thought that the cell nucleus contained the molecules that determine cell life.

The genes responsible for cell movement are therefore of utmost importance. In recent years, scientists have also shifted their research to the exterior of the cell membrane.

Please answer whether the following statements are true (T) or false (F) **(5 points)**

I. All epithelial cells are polar and essentially identical, with their basal surfaces attached to the basement membrane and closely adjacent to other cells.

The surface of the tissue, especially the top, is exposed.

II. Unlike typical ion channels, individual gap-junction channels, once formed, ...

It will remain open indefinitely.

III. Integrins can convert mechanical signals such as tension into molecular signals.

IV. The extracellular matrix, located outside the cell membrane, is a scaffold structure that stabilizes tissues.

It has little effect on the development, migration, diffusion, shape, and metabolism of neighboring cells.

V. In humans, all forms of fibronectin are formed by alternating splicing of the same long gene.

It comes from (alternative splicing).

I	II	III	IV	In

PART D: Fill in the blanks (15 points, 3 points per blank)

1. When slime molds encounter nutrient deficiency in their environment, their cells produce and release cAMP into the surrounding environment.

Attracting other slime molds to aggregate, eventually forming fruiting bodies, please answer the following question:

cAMP is produced by which enzyme? And is it broken down by which enzyme?

2. When animal cells are subjected to abiotic stress, the intracellular calcium ion concentration increases. Which intracellular protein can...?

It binds to calcium ions and subsequently initiates the activity of downstream enzymes.

3. On the cell membrane of animal cells, receptor proteins that can receive external signals can be roughly divided into three types, except for...

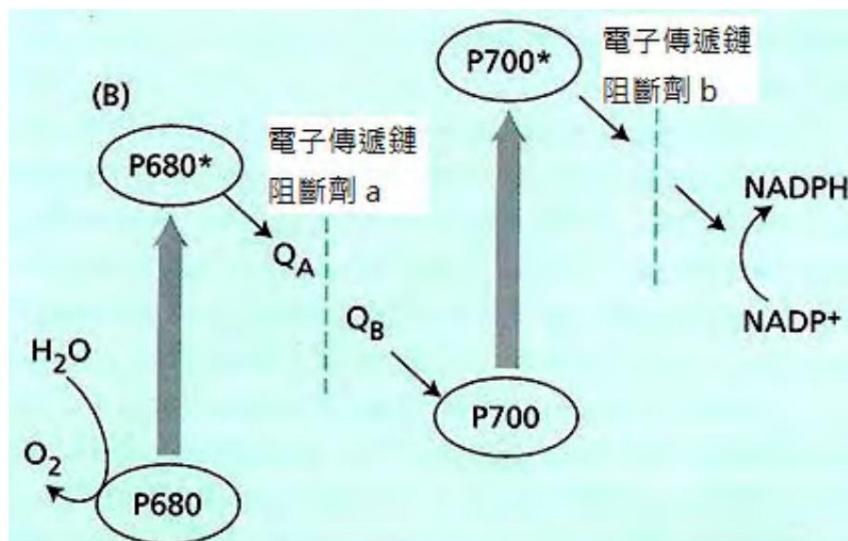
Besides the receptor proteins with enzyme activity (1), what other two are there?

Part Two: (100 points)

The following are topics related to plant physiology and morphology. Please answer the questions according to the descriptions for each topic.

I. Photosynthesis (5 questions, 25 points total)

Botanists often use electron blockers in the electron transport chain to study related issues of photosynthesis, including Optical System I (PSI), Optical System II (PSII), and Light Harvesting Complex I (LHCI) and Light Harvesting Complex II (LHCII). The physiological functions of these substances, and even the application of this knowledge in agricultural technology. For example, electron blocker a in the image below. By using the inhibitors b to block different positions in the electron transport chain, experimental designs can be employed to obtain light... The physiological functions and mechanisms of action of weed control. Applying this knowledge to agricultural technology can lead to the development of weed control methods. Emerging pesticides such as pesticides can enhance crop production potential.



The following questions relate to the electron transport chain in photosynthesis. Please answer the following sets of questions based on this information:

1. The energy required to excite one mole electron in photosystem I and photosystem II is 171 and 175 kJ/mol photons, respectively, and a pair of mole electrons can generate one mole of NADPH and two moles of ATP.

In terms of the energy transfer efficiency of NADPH and ATP formation, the theoretical energy of the photosystem electron transport chain...

Which of the following values is closest in terms of quantity transfer efficiency? (2 points)

(Note: 218 kJ/mol is required to form NADPH, while the free energy of $ATP \rightarrow ADP + P_i$ is -7.3 kcal/mol, 1 cal = 4.184 J).

- (A) 90%
- (B) 70%
- (C) 60%
- (D) 40%
- (E) 30%

answer: _____

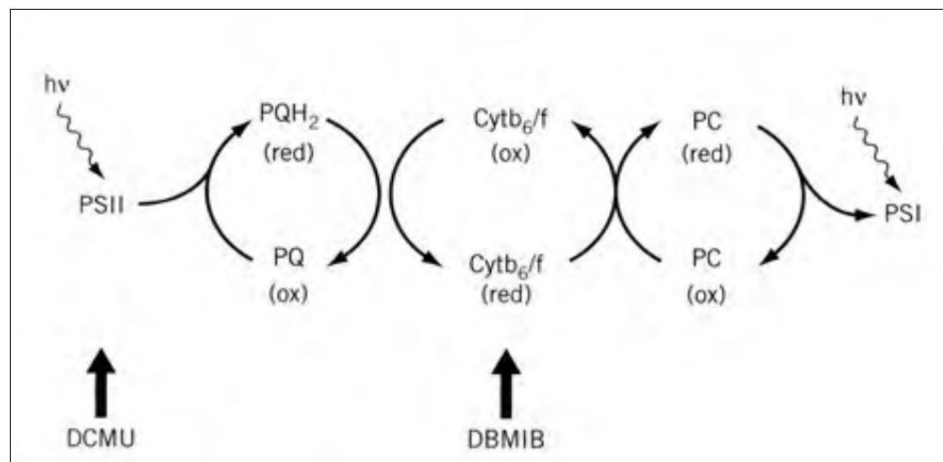
2. Please list the calculation formula (calculation process) and calculation result for the above question (8 points):

Answer:

3. What is the theoretical energy transfer efficiency of a cyclic electron transport chain? Please list the calculation formula (calculation flow) and...

Please calculate the results and compare the theoretical energy transfer efficiencies of the cyclic and non-cyclic electron transport chains, and their physiological significance. (5 points) Answer:

4. Botanists know that plants can adjust and alter their physiological activities according to their redox state (PQ) in the light response, including fine-tuning the structure of light-harvesting systems and increasing or decreasing the activity of metabolic enzymes, in response to changes in redox potential caused by changes in the external environment. DCMU and DBMIB in the diagram below are both electron blockers in the electron transport chain, with the blocking positions shown in the diagram. Under illumination, which electron blocker will inhibit chlorophyll synthesis in algae? What is the reason? (5 points)



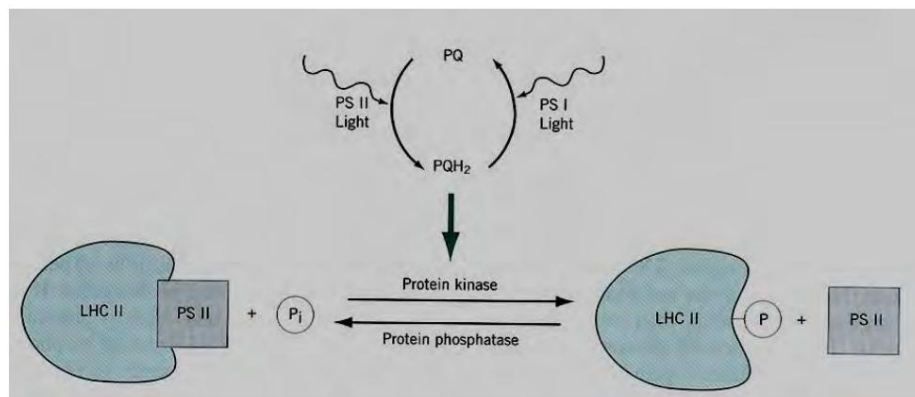
answer:

5. The diagram below illustrates the function of the fine structures regulating light in the light-harvesting system of plant leaves.

The binding and dissociation of PSII and LHCII are regulated by the PQ redox potential of the light-dependent reaction.

The PQ redox potential affects the activity of protein kinases and protein phosphatases, thereby regulating the binding and dissociation of PSII and LHCII. The mechanism of action is shown in the diagram below. Under light and in darkness, do PSII and LHCII bind or dissociate? What are the reasons? What is the physiological significance of this phenomenon?

(5 points)



answer:

II. Message Transmission (3 questions, **25 points in total**)

In 1992, botanists used tomato seedlings with two leaves as experimental material and discovered that mechanical damage to the tomato cotyledons triggers an action potential signal, causing inhibition of systemic proteinase in the distal first leaf (mature leaf), thus demonstrating the plant's defense mechanism. Two electrodes were attached to the stem at the base of the cotyledons and the petiole of the first leaf, respectively, to record the action potential. Based on this, please answer the following questions:

1. How do you think this botanist would design an experiment to prove that this action is a signal transduction generated by the action potential at the injured site, and to rule out that this action is not a familiar biochemical signal transduction pathway in plant defense? (**10 points**) Answer:

2. What do you think the graph of the action potential measured by this botanist looks like? Please draw and label the important characteristics of the action potential: (5 points)

Answer:

3. What does the "plant defense pathways of biochemical signaling that we are familiar with" refer to in the above question? Please explain. (10 points) Answer:

III. Cell Specialization (2 questions, **25 points total**)

The apical meristem of angiosperms typically exhibits distinct stratification. L1 (or the outer layer, also known as tunica) primarily undergoes anticlinal division to form the epidermis; while the inner layers (L2 in maize; L2 and L3 in Arabica mustard) constitute the corpus of the meristem. During epidermal development, protoderm cells primarily undergo anticlinal division to form a single-layered epidermis. By studying mutant strains with defects in epidermal cell division and differentiation, one can visualize cell division and differentiation...

The mechanism of mutual coordination is crucial. Mutations in the multicellular layer (Xcl1) cause oblique periclinal division in prodermal cells , resulting in a multi-layered epidermal morphology. Dosage analysis of the Xcl1 mutation shows that this mutation leads to overexpression of the normal gene, resulting in an overdose of XCL1. Increasing the mutation in XCL1 prolongs cell division during leaf development. As a result, cells that have received the message to specialize into epidermis continue to divide into multiple layers and specialize according to the relationship between sister cells, rather than according to the location of the cell.

In addition to causing a distinct multilayered epidermis in leaves, the Xcl1 phenotype also affects the apical meristem of the stem . The apical meristem of Xcl1 is typically 35% shorter than that of normal stems, and its newly formed leaf primordium involves more cells. This phenomenon suggests that Xcl1 may not only be related to the division and specialization patterns of the primordium, but also play an early role in regulating meristem size and the initiation of leaf development. It may also provide connections between L1 cell division patterns, hormone transport pathways, and structural characteristics such as meristem maintenance.

Figure 1 shows the phenotype of the Xcl1 mutation in maize. As shown in Figures 1A and 1B, *the third leaf of the wild-type completely covers the meristem, while the third and fourth leaves of the Xcl1 type fail to do so* , indicating that the Xcl1 leaves are narrow from early development. Before the second leaf, *the phylogenetic morphology of the wild-type and Xcl1 types is very similar, although the apical dome of the Xcl1 type* is shorter than that of the wild- type (Figures 1C and 1D). The most prominent phenotype of Xcl1 is its multi-layered epidermis, resulting from oblique division of L1 during early leaf development (Figure 1H). Furthermore, Xcl1 leaves contain a smaller number of intermediate veins (Figure 1I).

Based on the above explanation and the results in Figure 1, answer the following questions (1)-(2).

1. Referring to the text description below Figure 1, clearly indicate (or circle) the location of the corresponding structures in each small diagram (AH) of Figure 1, and write the name of the structure in the blank space on both sides of the diagram (as shown in Figure 1A). (2 points for each correct labeling; maximum 15 points)

Meristematic tissue, meristem dome, first leaf, second leaf, third leaf, single epidermis, multilayered epidermis, vascular bundle

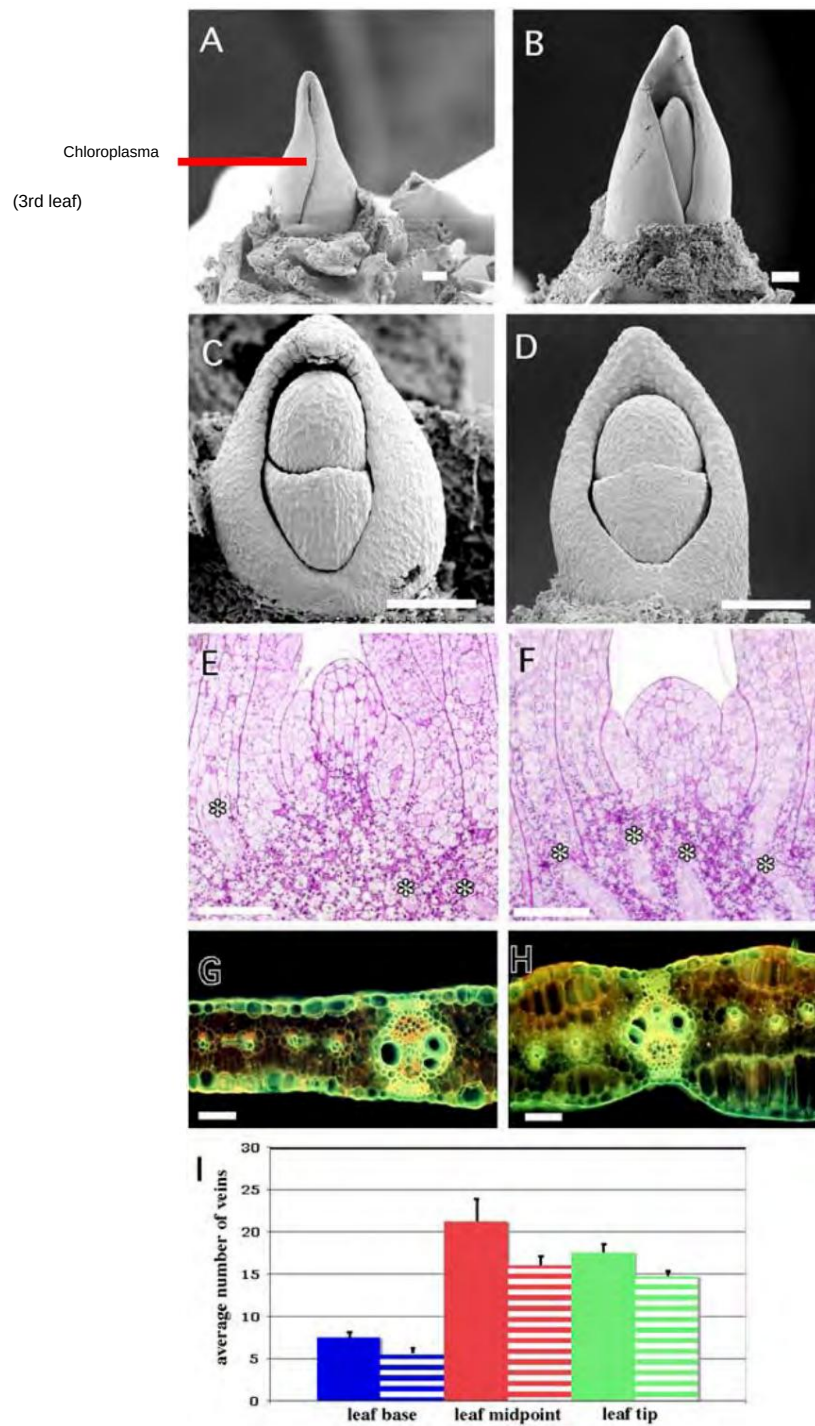


Figure 1. Meristems and leaves of wild maize plants (A, C, E, and G) and Xcl1 mutants (B, D, F, and H).

Primogeneous organism. A, Meristem of the wild plant, where the third leaf completely encloses the meristem and the edges overlap. B, Meristem of Xcl1, where the third leaf does not overlap. C, Meristem of the wild plant, containing 1-2 leaves. D, Meristem of Xcl1, containing 1-2 leaves and showing a short, slightly domed meristem. E, Meristem of the wild plant.

The longitudinal section shows the formation of vascular bundles in the primordia of the second and third leaves. The longitudinal section of the F, Xcl1 meristem...

The vascular bundles extend relatively deep into the meristematic tissue. G, Cross section of a wild plant leaf. H, Leaf of Xcl1.

A cross-section reveals multiple layers of epidermis. Three different parts of the leaves from plant I (solid stripes) and Xcl1 (striped) are also visible.

The number of leaf veins (base, midpoint, tip).

2. Based on the above research, extend the explanation of leaf development, that is, from the perspective of cell division and specialization, explain the complete formation process from the phylogenetic stage to the mature leaf, especially how L1 and L2 in the phylogenetic stage divide and specialize into various constituent tissues within the leaf. (10

points) Answer:

IV. Artificial Breeding (2 questions, **25 points in total**)

Horticultural flowers are often produced through artificial breeding. The following four photos (AD) show flowers that look similar. Their common names and classifications are as follows: Dahlia (Asteraceae), Rieger Begonia (Begoniaceae), Rose (Rosaceae), and Double-flowered Hibiscus (Malvaceae). **1.**

(Next Page) Please label the flower in the picture

below it (1.5 points per box) and explain your identification.

Characteristics of the genus and family (1.5 points per blank). (Total 12 points)



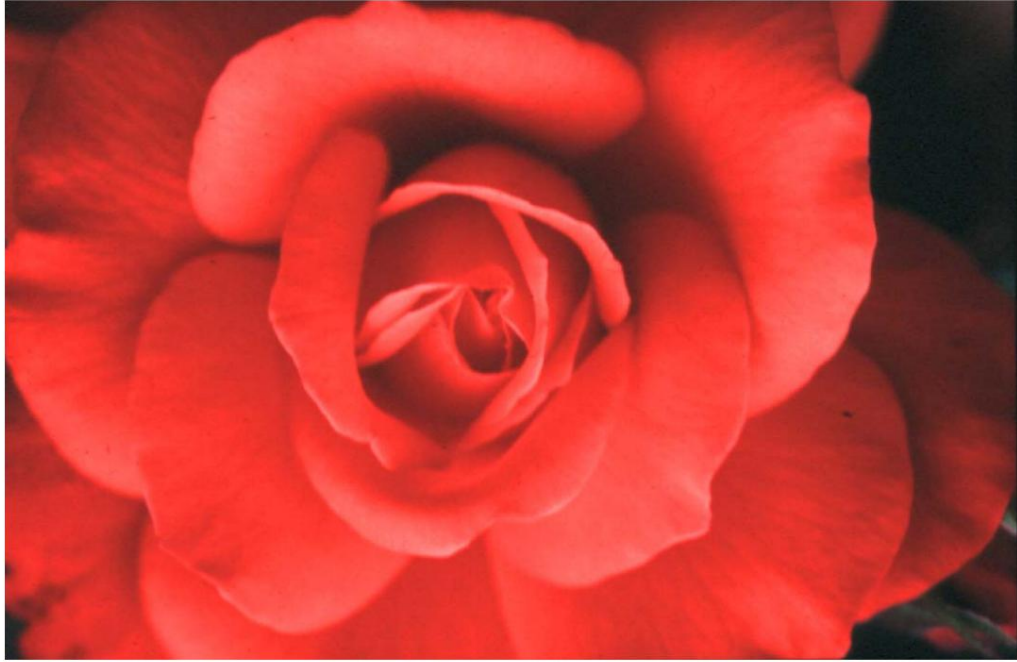
(A) Name:

Family characteristics:



(B) Name:

Classification characteristics:



(C) Name:

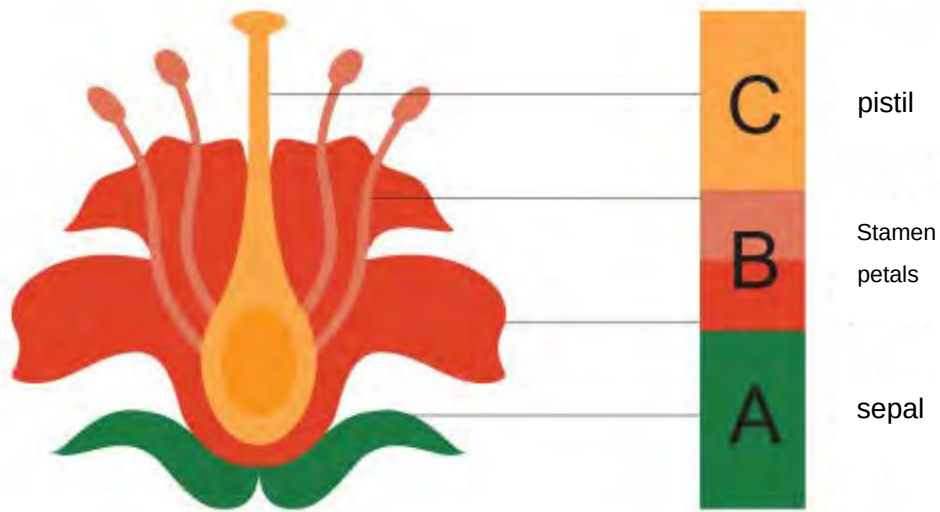
Family characteristics:



(D) Name:

Family characteristics:

2. Referring to the ABC model of flower development shown in the diagram below, what mutations do you think might have occurred in the above flowers? How can these mutations be screened out from the wild type? And how can these flowers be cultivated to maintain their characteristics? (13 points)



Answer: (1) Mutation type (6

points) Flower commonly known as mutation	

(2) Mutation screening methods (4 points)

(3) Cultivation methods (3 points)

Part Three: 100 points

I. Short Answer Questions: 25 points

1. The respiratory quotient (RQ) is defined as the ratio of the amount of carbon dioxide exhaled to the amount of oxygen consumed by an animal during respiration. (4 points total)

(a) Calculate the reaction equation $C_xH_yO_z + (x + y/4 - z/2) O_2 \rightarrow xCO_2 + (y/2) H_2O$, using the carbonization process.

Find the RQ value when the compound is the main energy source (2 points, show the calculation process).

(ii) Explain why the RQ value of humans is higher during intense exercise than during rest (2 points).

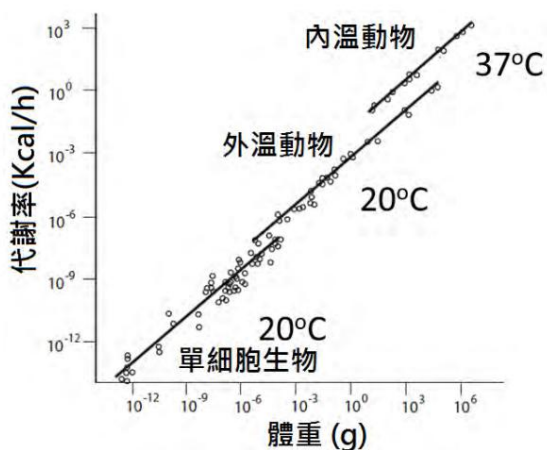
2. The metabolic rate of an organism is linearly related to the 0.75th power of its body weight (as shown in the figure below). (6 points in total)

(i) Write down the relationship between the metabolic rate per unit body weight and body weight (2 points, please show the derivation process).

(ii) Which has a higher metabolic rate per unit body weight, a mouse or an elephant?

Write down possible biological explanations (2 points).

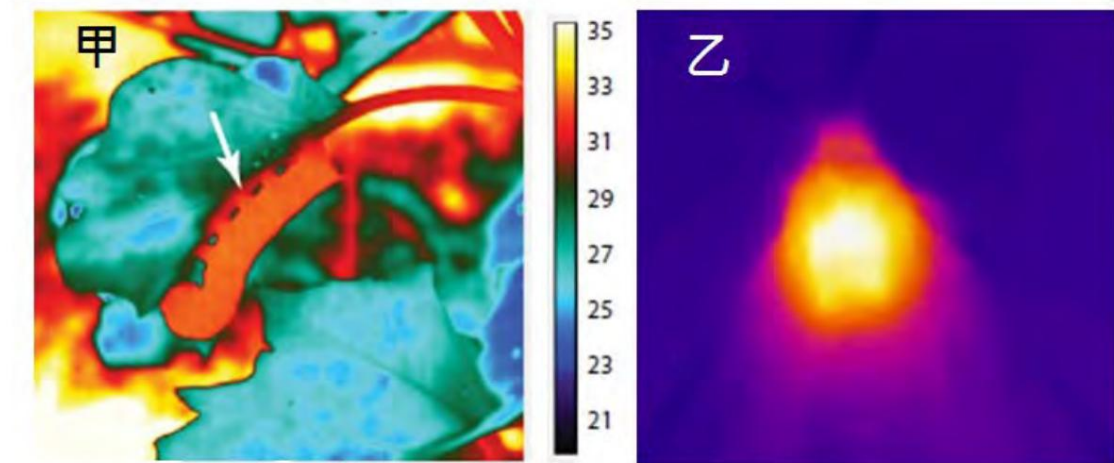
(i) Based on the above information and the relationship between body size and surface area, explain why endothermic animals have a lower limit to their body size. (2 points)



Number» 3. Ephemeral animals can regulate their body temperature through physiological or behavioral means. Figures (A) and (B) below show measurements taken with a thermal imager.

Insect larvae during the day (Figure A) and the temperature of their grazing food, and the body temperature of a moth (Figure B) before taking flight at night.

And ambient temperature. (Total 15 points)



(a) (a) What is the most likely way for larvae to raise their body temperature through heat exchange with the outside world ? (2 points)

What is the main external heat source? (2 points)

(ii) Several locations on the larvae's body wall in Figure A that have lower body temperatures (indicated by arrows) may contain what kind of structures?

Identify the location (name it, 2 points) and provide an explanation to support your hypothesis (2 points).

(III) Explain the physiological mechanism by which the moths in Figure (B) raise their body temperature. (2 points)

(iv) Of the two different modes of body temperature regulation mentioned above, which consumes more energy from the organism, and why? (2 points)

(V) Please explain (1) the effect of temperature on the water vapor content in the air (1 point) (2) the effect of temperature on the metabolic rate of organisms.

(1 point) What are the effects of increasing body temperature on water loss from the body surface of organisms? (3)

II. Fill in the blanks (25 points)

1. Urine formation is mainly determined by three processes: filtration (A), re-filtration (B), and urination (C).

Absorption (B) and secretion (C) are the mechanisms by which substances are excreted in urine. The amount of a substance excreted in urine can be calculated using the above three parameters and a formula.

Indicates: (1) _____ (2 points)

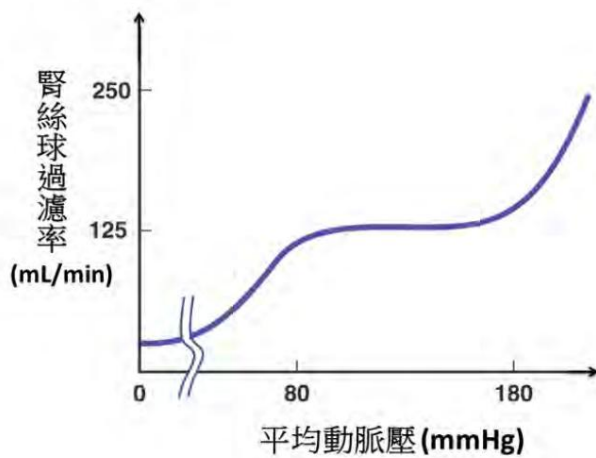
2. The filtration efficiency (effective filtration pressure) of the glomerulus can be determined by four pressures. Please list the four pressures.

Pressure (2), (3), (4), (5) (2 points each), and how these four pressures determine the effective filtration pressure (6) (3 points)

3. If kidney disease occurs and blood proteins enter Bowman's capsule, what pressure will increase? (7) (2 points) 4. The horizontal axis of the following diagram represents mean arterial pressure (mmHg), and the vertical axis represents glomerular filtration rate.

Glomerular filtration rate (ml/min); Please briefly describe the relationship between them (8 points) (5 points) _____

(Stanfield, Principles of Human Physiology, 5th Ed.)



5. The so-called filter load (or filtration capacity) refers to the amount of a specific solute filtered per unit time.

Quantity. Continuing from the previous graph, in hypertensive patients with a blood pressure of 200 mmHg, the GFR is 250 ml/min; glucose

If the concentration of glucose in plasma is 1 mg/ml, then the filtration load of glucose is (9) (5 points). _____

Answer bar

(1)			
(2)	(3)	(4)	(5)
(6)			
(7)	(8)		
(9)			

III. Short Answer Questions (25 points)

1. Mr. Wang was stung by a jellyfish while snorkeling in the sea. Although he swam back to shore immediately, he quickly suffocated due to respiratory failure. Researchers captured the jellyfish and isolated toxin A from the stinging cells of its tentacles, quickly confirming that this toxin was the cause of Mr. Wang's suffocation due to the dysfunction of his respiratory muscles. Toxin A is a large protein that cannot cross the blood-brain barrier. Please describe the functioning of respiratory muscles (5 points), and based on this, list five possible mechanisms by which toxin A causes dysfunction of these respiratory muscles and explain the reasons (20 points).

IV. Short Answer Questions (25 points)

Please choose 10 animals that you eat in daily life, plus one outgroup, and draw a diagram based on their morphology.

Based on the most minimal method, the kinship hypothesis is reconstructed and fully resolved. The following information is provided:

(1) Why was this animal chosen as the outgroup? (5 points)

(2) If these selected animals are all considered as an operational taxonomic unit (OTU), then what are the common characteristics of each monophyletic group in the phylogenetic relationships formed by these OTUs?

Why is (synapomorphy)? (15 points)

(3) Please select five co-derived traits and explain their ecological significance, as well as whether these traits might contribute to this.

What accounts for the evolutionary success of the group represented by this organism? (5 points)

Theoretical Assessment Questions for the 2016 International

Biology Olympiad

National Team Selection **Camp**

Part Four

Student Code **Y**A1



Precautions:

1. This section of the exam consists of 4 pages (excluding the cover) and is worth 100 points. Please check for any missing pages.

Please inform the test administrator immediately if any pages are missing.

2. When answering the questions, please check that the cover of the test paper matches your test number.

3. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.

4. You may write your answers on the back of the questions, but please indicate the question number.

Part Four: (100 points)

ÿ PART A (Total 50 points)

In the 1970s, DNA could be sequenced using two methods: one was the chemical cutting method invented by Allan Maxam and Walter Gilbert, which was more complicated; the other was a simpler and more widely used method...

The dideoxynucleotide chain termination method was invented by Frederick Sanger. Furthermore, improvements in nucleic acid colloidal electrophoresis have enabled the differentiation of highly different single-base nucleotides.

The Sanger sequencing method utilizes tiny DNA fragments to sequence hundreds of nucleotides at once.

Using enzymes as a template based on the DNA to be sequenced, radioactive primers labeled with radioisotopes are synthesized.

Isotope-labeled complementary DNA fragments were added to reaction tubes in four groups: A, C, G, and T.

In addition to four basic nucleotides (dNTPs) as raw materials, a small amount of dideoxynucleotide (ddNTP) was added to each.

The following

groups were selected: Group A:

dNTPs + ddATP; Group C: dNTPs +

ddCTP; Group G: dNTPs + ddGTP;

Group T: dNTPs + ddTTP

After the reaction is complete, the reactants are injected into electrophoretic colloidal sample cells A, C, G, and T, respectively.

After electrophoresis, the colloid is dried and then radiographically developed using X-ray film to obtain the sequencing reaction product, DNA.

Image of the fragment. Xiaoying wants to use the Sanger method with radioactive isotope-labeled primers *5'-----3' to sequence a DNA

fragment with the sequence in parentheses (-----ACGACGCAGGACATTAGAC), where...

The dashed line represents the sequence complementary to the initiator sequence. The image below shows the radioisotope imaging of its sequencing results.

However, only the image bands of DNA fragments from group T are shown. Please select the groups C, A, and G from the image.

The radioisotope imaging bands of the DNA fragments from the sequencing reaction products are marked below each group in the figure.

(18 points)

The nucleotide bases added in groups 1 and 2 are: 1. dNTPs + ddCTP and ddATP.

2. dATPÿdGTPÿdTTP ÿ ddGTP

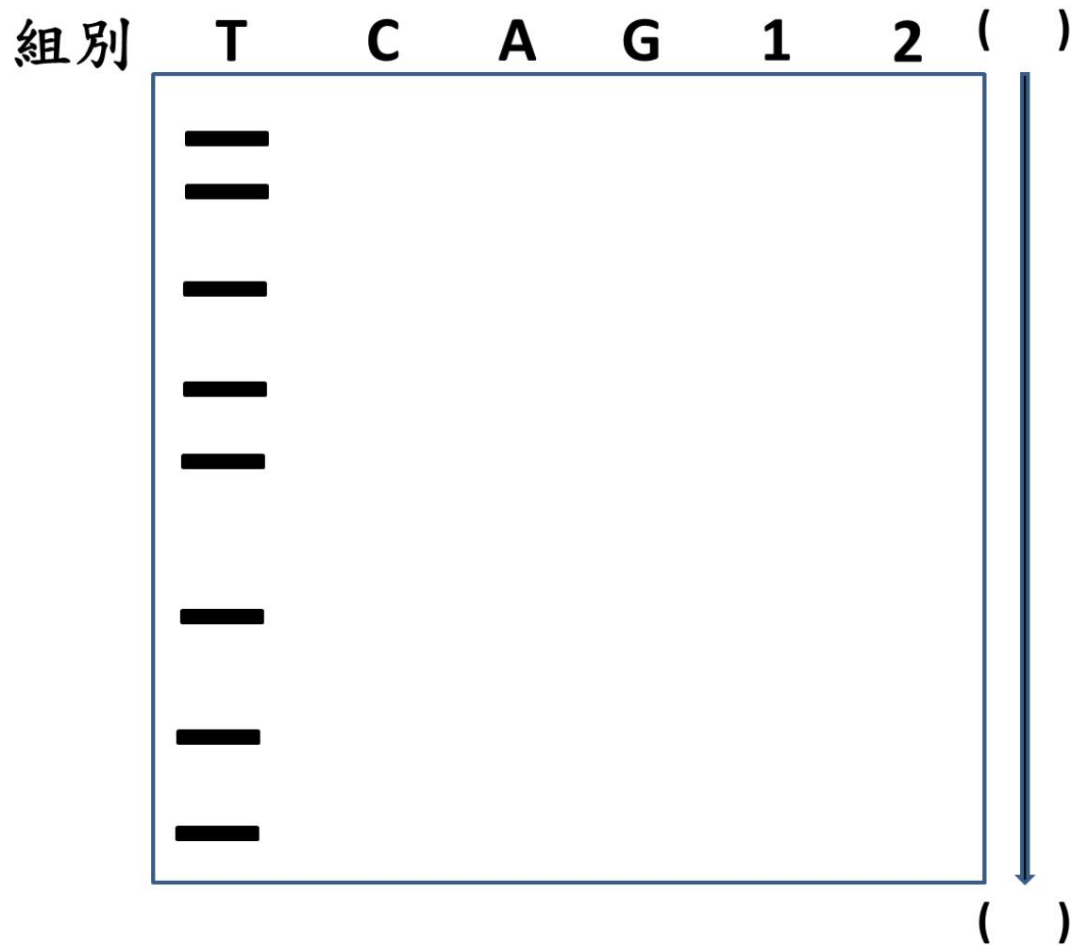
Please also indicate the radiographic bands of the DNA fragments from the reaction product in the figure below each group.

(12 points)

Please answer the following questions: **1.**

The arrow on the right side of the diagram indicates the direction of DNA fragment migration in the sequencing reaction. Please mark the direction in the parentheses at both ends of the arrow.

Show the symbols for the positive and negative electrodes. (5 **points**)



2. What type of electrophoretic gel is used for DNA sequencing? (5 points)

3. What enzyme is used in the Sanger sequencing method? Please explain its relationship to DNA replication in living cells.

Are there differences in the enzymes used? (5 points)

4. Please explain whether the primers used in the Sanger sequencing method are different from those used in DNA replication in living cells. (5 points)

PART B (3 questions, 50 points total)

- 1.** G6PD deficiency in humans is caused by glucose-6-phosphate dehydrogenase (G6PD).

Phosphate-dehydrogenase deficiency is an X-linked genetic disorder. [The enzyme is involved in the action of this enzyme.]

The substance is _____⁹ _____⁹

- 2.** Patients may experience hemolysis, jaundice, or acute renal failure upon exposure to camphor, broad beans, and certain medications.

The reason is that the main product of its biochemical reaction is NADPH, which can assist in antioxidant activity; a lack of this enzyme...

In individuals with abnormal blood composition, red blood cells are more susceptible to oxidative damage, leading to hemolysis. Please list three methods to confirm this in a person.

Does this hereditary disease exist? (5 points for each method, 15 points in total)

- 3.** Scientists conducted tests on 1,000 people each in northern, central, and southern Taiwan, finding that in the north...

The prevalence of G6PD deficiency is 3.06%; 2.98% of the population in central Taiwan has G6PD deficiency; and in southern Taiwan...

They make up 2.96% of the total population. What is the incidence of G6PD deficiency in Taiwan's population?

(10 points)

4. Assuming that the total number of male and female students in Taiwan is roughly the same, please estimate what percentage of patients are female?

(25 points)

Part Five: (100 points)

I. Multiple Choice Questions (4 questions in total, **24 points in total**)

- _____ 1. Ecotypes refer to a group of plants that exhibit characteristics adapted to the environment of their growing region.

A species possesses distinct genotypes and morphologies. Ecotypes are the result of genetic variation and natural selection,

representing different genotypes. Therefore, even when transplanted to the same habitat, they maintain their stable differences.

Ecotypes are precursors to new species. Which of the following characteristics is most likely to prevent a species from developing a new ecotype?

(A) High dispersal ability (B)

Long generation time (C)

Competition for limited resources

(D) Stable habitat conditions

(E) High overlap of environmental conditions

- _____ 2. A type of grass contains two ecotypes: a heavy metal-tolerant ecotype and a heavy metal-intolerant ecotype.

However, the offspring produced after hybridization have very low adaptability in their original habitat. Under these conditions, for these two ecotypes to maintain their genetic consistency, which of the following is most relevant?

(A) Pollen dispersal capacity of the two ecotypes

(B) Number of hybridizations between the two ecotype

populations (C) Growth status of each ecotype population

(D) The number and reproductive capacity of hybrids from the two ecotypes

(E) The stability of the native environmental conditions of the two ecotypes

- _____ 3. Which of the following characteristics best represents a climax community?

(A) High species richness

(B) High species equitability

(C) High productivity

(D) More dominant species

(E) Low species turnover rate

- _____ 4. It is generally believed that societies in the later stages of succession will be limited in their access to nutrients. What are the possible reasons for this?

(A) This successional stage is highly competitive for nutrient

resources. (B) The rate of weathering decreases, thus limiting nutrient

availability. (C) The rate of primary production limits the energy required

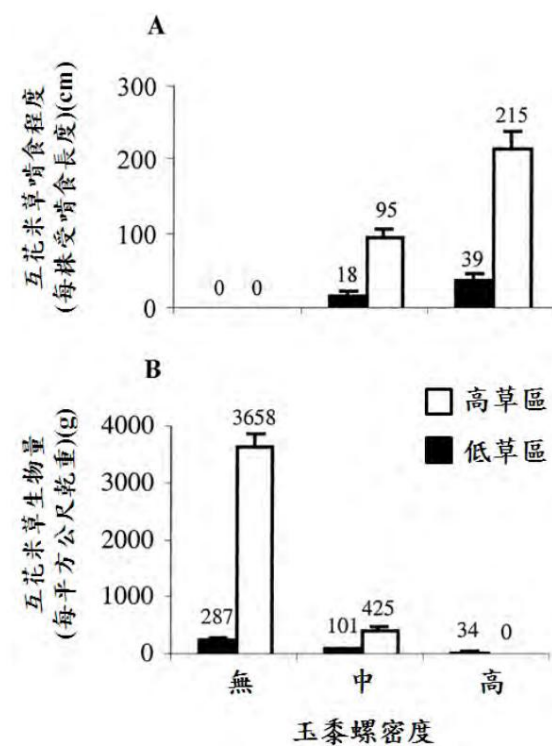
for nutrient acquisition. (D) Fewer invasive alien species provide less nutrient

supply. (E) Over a long successional stage, nutrient loss exceeds replenishment.

II. True or False Questions (12 questions, **24 points in total**)

Spartina alterniflora is a dominant plant in the intertidal mudflats of the Atlantic coast of North America. In Georgia, USA, *Spartina alterniflora* exhibits two distinct growth forms depending on the nutrient concentration of the mudflats: in lower-lying swamps with higher nitrogen concentrations, it grows to 200-300 cm (tall grass zone); in higher-lying swamps with lower nitrogen concentrations, it only reaches 40-80 cm (low grass zone). The local corn snail is also mentioned.

*Littoraria irrorata** feeds on the dead leaves of **Spartina alterniflora**, but may also feed on the fresh leaves. The normal density of **Littoraria irrorata** in this area is approximately 600 individuals per square meter, with a lifespan of three years. To investigate the effects of animals and nutrients on **Spartina alterniflora** growth, Silliman and Bertness (2002) conducted a series of operational experiments. They enclosed three types of sample areas in both tall and low-grass areas using netting. Within these 1 m² sample areas, they established three different adult snail densities: high (1200 individuals), medium (600 individuals), and none (0 individuals). All other environmental conditions were kept constant, and animals were not allowed to enter or leave these netted sample areas (a total of 48 sample areas, 2 types of **Spartina alterniflora** sample areas × 3 species of snail density × 8 replicates). Eight months later (October of that year), it was found that the density of *Spartina alterniflora* in the enclosure area had hardly changed. The degree of grazing on *Spartina alterniflora* (the length of the grazing leaf per plant) was measured, and the upper part of *Spartina alterniflora* was collected, dried, and the biomass was measured (see figure below).



Note: Image modified from Silliman, BR and MD Bertness. 2002. A Trophic Cascade Regulates Salt Marsh

Primary Production. Proceedings of the National Academy of Sciences (USA) 99: 10500-1050

Which of the following statements is true or false? Please indicate with T (True) and F (False).

_____ (A) The corn snail feeds on the fresh leaves of *Spartina alterniflora*.

_____ (B) Within the netted sample area, only the corn snail feeds on the fresh leaves of *Spartina alterniflora*.

_____ (C) The extent to which *Spartina alterniflora* was eaten by *Cinnamomum campestris* was unrelated to nutrient concentration.

_____ (D) In areas with higher nutrient levels, the secondary production of corn snails should also be higher.

_____ (E) In areas with higher nutrient levels, each corn snail consumes an average of higher biomass of *Spartina alterniflora*.

_____ (F) As the density of **Spartina alterniflora** increased, the average biomass of **Spartina alterniflora** consumed by each snail also decreased significantly.

Low

_____ (G) The level of nutrients affects the biomass of *Spartina alterniflora*.

_____ (H) The density of **Spartina alterniflora** affects its biomass.

_____ (I) The higher the density of *Spartina alterniflora*, the higher the primary yield of *Spartina alterniflora*.

_____ (J) In areas with higher nutrient levels, the net primary production of *Spartina alterniflora* is also higher.

_____ (K) If natural enemies of the corn snail can enter the sample area, the biomass of *Spartina alterniflora* may change significantly.

_____ (L) In sample areas with *Spartina alterniflora*, the amount of primary production of *Spartina alterniflora* consumed by *Spartina alterniflora* does not exceed one-third.

Half

III. Short Answer Questions (2 questions, **52 points in total**)

1. The main content of ecology is to explore the interaction between species and the environment, including how the environment...

Influences on species abundance and distribution. To understand how the environment influences species abundance and distribution, there are several...

The research methods employed are extensive, including direct field observation, hands-on experiments, and mathematical simulations.

Methods. Describe the situations in which each of these three methods is most suitable, and also explain the advantages and disadvantages of each method.

What are the advantages and disadvantages? (25 points)

2. The burrowing owl is a small owl common in the American prairie, feeding on small animals, including vertebrates and...

An invertebrate. It uses abandoned burrows of other burrowing animals in the grasslands as homes and breeds its young.

Researchers observed that it collects cattle dung on the grasslands and places it near its burrows. Suggest possible reasons.

Hypothesize to explain this phenomenon and design experiments to verify the proposed hypothesis. Also, briefly describe its relationship with cattle.

A type of relationship. (27 points)